

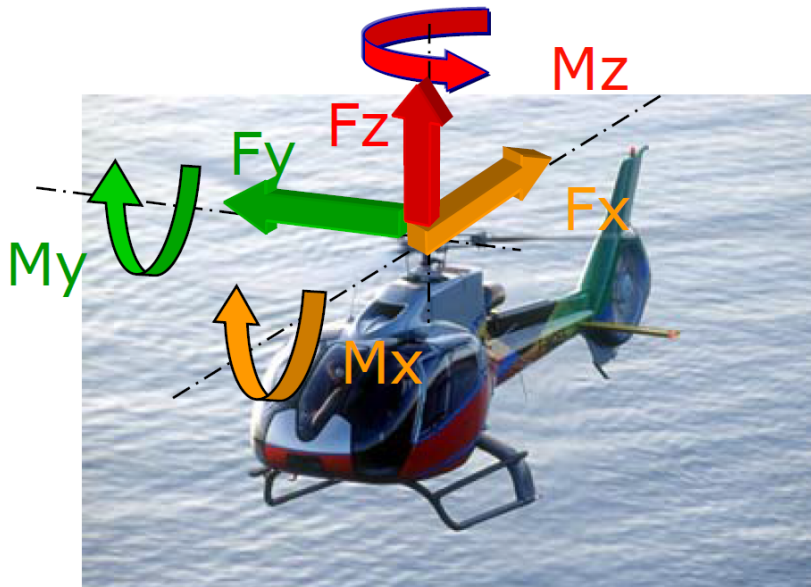
Helicopter

3 degree of freedom (dof)

helicopter longitudinal model

Equations of motion in a 3-dof model

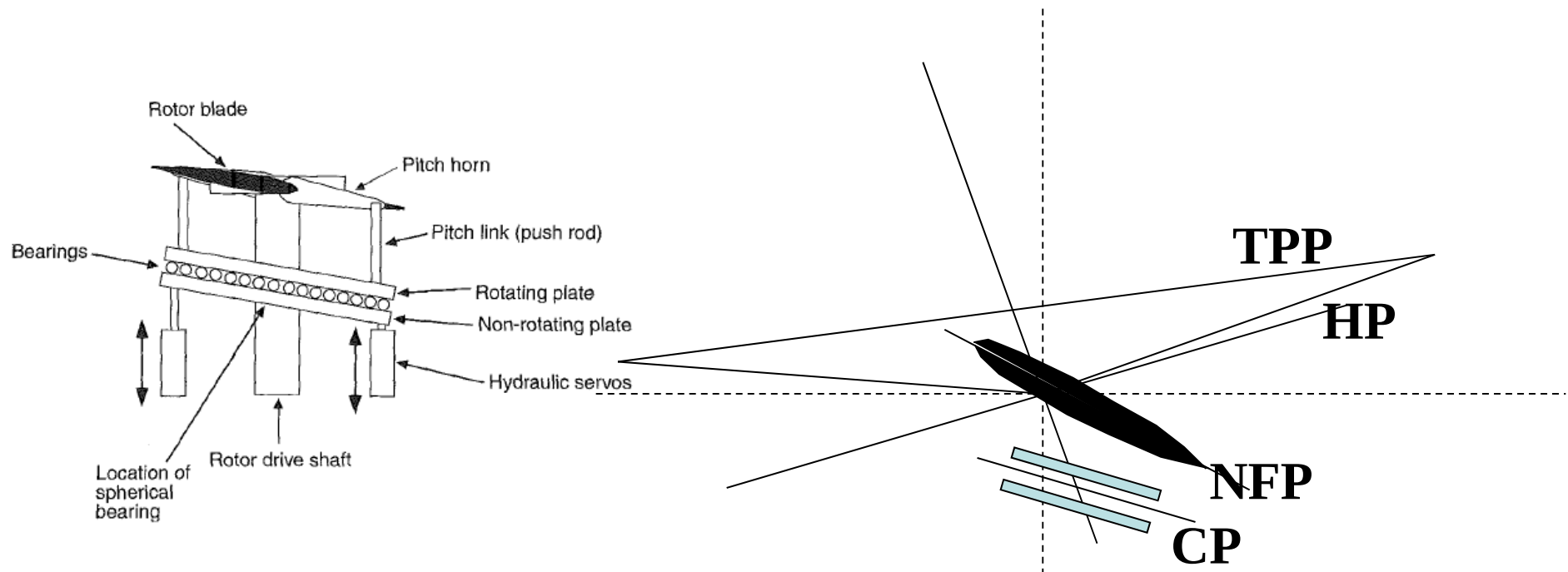
Dynamic equations of motion in a longitudinal plane are the fundamental equations of dynamics:



$$\left\{ \begin{array}{l} \dot{u} = -qw + \frac{F_x}{m} \\ \dot{w} = qu + \frac{F_z}{m} \\ \dot{q} = \frac{M_y}{I_y} \\ \dot{\theta}_f = q \end{array} \right. \quad (1)$$

We have thus to find the forces and moments acting on the helicopter in the body system of reference:

Main rotor reference systems



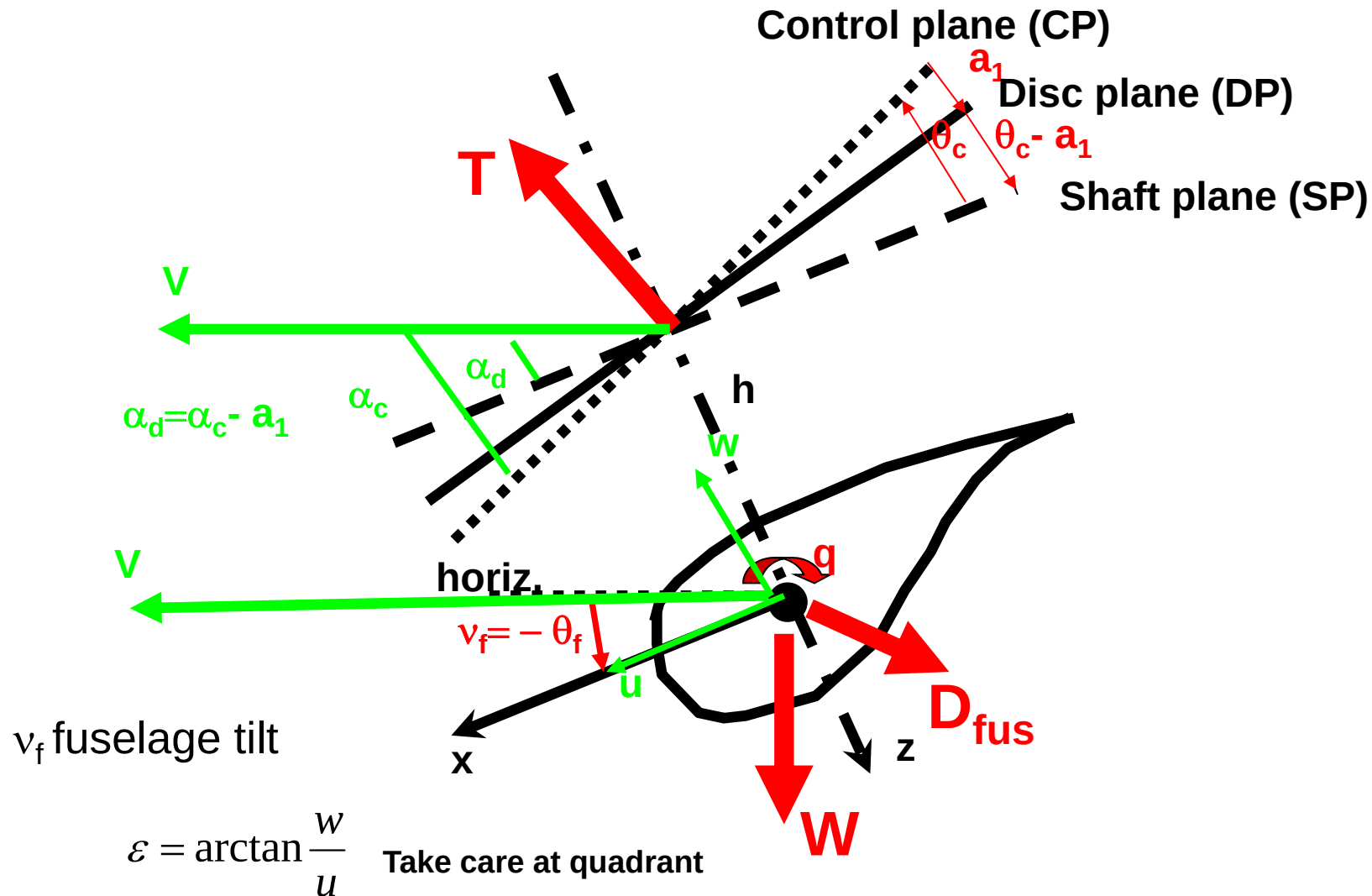
TPP (DP) = tip-path-plane (disc plane) is *the plane formed by the blade tips*

SP (HP) = shaft plane (hub plane) is *the plane perpendicular to the hub*

NFP = no-feathering plane is *the plane where an observer sees no variation in cyclic pitch*

CP = control plane (swashplate plane) is *the plane containing the swashplate*

Equations of motion in a 3-dof model



Equations of motion in a 3-dof model

Assuming Ox axis parallel with the helicopter shaft plane system and neglecting the rotor drag forces ($H_i=0$) gives:

$$\begin{aligned} F_x &= -W \sin \theta_f - D_{fus} \cdot \cos \theta_f + T \sin(\theta_c - a_1) \\ F_z &= W \cos \theta_f - D_{fus} \cdot \sin \theta_f - T \cos(\theta_c - a_1) \\ M_y &= -Th \sin(\theta_c - a_1) \end{aligned} \quad (2)$$

But, the fuselage pitch angle can be expressed as:

$$\cos \theta_f = u / V \quad (3)$$

$$\sin \theta_f = w / V \quad (4)$$

Substituting (2) –(4) into (1) gives:

Helicopter equations of motion in a 3-dof longitudinal model

$$\begin{aligned}\dot{u} &= -g \sin \theta_f - \frac{D_{\text{fus}}}{m} \cdot \frac{u}{V} + \frac{T}{m} \sin(\theta_c - a_1) - qw \\ \dot{w} &= g \cos \theta_f - \frac{D_{\text{fus}}}{m} \cdot \frac{w}{V} - \frac{T}{m} \cos(\theta_c - a_1) + qu \\ \dot{q} &= -\frac{T}{I_y} h \sin(\theta_c - a_1) \\ \dot{\theta}_f &= q\end{aligned}\tag{5}$$

Remember: Thrust force

$$C_{T\,elem} = \frac{C_{l\alpha}\sigma}{4} \left[\frac{2}{3} \theta_0 \left(1 + \frac{3}{2} \mu^2 \right) - (\lambda_i + \lambda_c) \right]$$

$$C_{T\,Gl} = 2\lambda_i \sqrt{\left[\frac{V}{\Omega R} \sin(\alpha_c - a_1) + \lambda_i \right]^2 + \left[\frac{V}{\Omega R} \cos(\alpha_c - a_1) \right]^2}$$

Building a flight mechanics model for piloted simulations

3-dof longitudinal model

3 degree of freedom longitudinal model

Course lecture notes
page 95, AE4-314

airplane parameters $M, I_y, C_D, S, \Omega, R$
control variables θ_0, θ_c
state variables u, w, q, θ_f
$V = \sqrt{u^2 + w^2}$ $\alpha_c = \theta_c - \arctan \frac{w}{u} \quad (\text{eventually } +\pi) !!$ $\mu = \frac{V}{\Omega R} \cdot \cos \alpha_c$ $\lambda_c = \frac{V \sin \alpha_c}{\Omega R}$
choose λ_1 until $F(\lambda_1) = 0$ $a_1 = \frac{\frac{8}{3} \mu \theta_0 - 2 \mu (\lambda_c + \lambda_1) - \frac{16}{\gamma} \frac{q}{\Omega}}{1 - \frac{1}{2} \mu^2}$ $C_{T_{HEM}} = \frac{1}{4} c_{la} \sigma \left[\frac{2}{3} \theta_0 \left(1 + \frac{3}{2} \mu^2 \right) - (\lambda_c + \lambda_1) \right]$ $C_{T_{Glow}} = 2 \lambda_1 \sqrt{\left(\frac{V}{\Omega R} \cos(\alpha_c - a_1) \right)^2 + \left(\frac{V}{\Omega R} \sin(\alpha_c - a_1) + \lambda_1 \right)^2}$ $F(\lambda_1) = C_{T_{HEM}} - C_{T_{Glow}}$
The following parameters are now known: λ_1, a_1 and C_T $T = C_T \rho (\Omega R)^2 \pi R^2$ $D = C_D \frac{1}{2} \rho V^2 S$ $\dot{u} = -g \sin \theta_f - \frac{D}{m} \frac{u}{V} + \frac{T}{m} \sin(\theta_c - a_1) - q w$ $\dot{w} = g \cos \theta_f - \frac{D}{m} \frac{w}{V} - \frac{T}{m} \cos(\theta_c - a_1) + q u$ $\dot{q} = -\frac{T}{I_y} h \cdot \sin(\theta_c - a_1)$ $\dot{\theta}_f = q$

$$(u, w, \theta_f, q) \Rightarrow (\dot{u}, \dot{w}, \dot{\theta}_f, \dot{q})$$

Initial data

airplane parameters $M, I_y, C_D, S, \Omega, R$
control variables θ_0, θ_c
state variables u, w, q, θ_f

Calculate parameters

$$V = \sqrt{u^2 + w^2}$$

$$\alpha_c = -\arctan \frac{w}{u} + \theta_c$$

If $u < 0$ add $\alpha_c + \pi$

$$\mu = \frac{V}{\Omega R} \cdot \cos \alpha_c$$

$$\lambda_c = \frac{V}{\Omega R} \cdot \sin \alpha_c$$

Enter in a loop to calculate λ_i, a_1 and C_T

Choose λ_i till $F(\lambda_i) = 0$

$$a_1 = \frac{-\frac{16}{\gamma} \frac{q}{\Omega} + \frac{8}{3} \mu \theta_0 - 2\mu(\lambda_c + \lambda_i)}{1 - \frac{1}{2} \mu^2}$$

$$C_{T_{\text{element}}} = C_{l_a} \frac{\sigma}{4} \left[\frac{2}{3} \theta_0 \left(1 + \frac{3}{2} \mu^2 \right) - (\lambda_c + \lambda_i) \right]$$

$$C_{T_{\text{Glauert}}} = 2\lambda_i \sqrt{\left(\frac{V}{\Omega R} \cos(\alpha_c - a_1) \right)^2 + \left(\frac{V}{\Omega R} \sin(\alpha_c - a_1) + \lambda_i \right)^2}$$

$$F(\lambda_i) = C_{T_{\text{element}}} - C_{T_{\text{Glauert}}}$$

Equations of motion

$$\dot{u} = -g \sin \theta_f - \frac{D}{m} \frac{u}{V} + \frac{T}{m} \sin(\theta_c - a_1) - qw$$

$$\dot{w} = g \cos \theta_f - \frac{D}{m} \frac{w}{V} - \frac{T}{m} \cos(\theta_c - a_1) + qu$$

$$\dot{q} = -\frac{T}{I_y} h \cdot \sin(\theta_c - a_1)$$

$$\dot{\theta}_f = q$$

$$T = C_T \rho (\Omega R)^2 \pi R^2$$

$$D = C_D \frac{1}{2} \rho V^2 S$$